Fiscal Year:	FY 2016	**Task Last Updated:**	FY 02/29/2016
PI Name:	Hasenstein, Karl Ph.D.		
Project Title:	Microgravity and Magnetic Fields - Directional Root Growth and Gene Expression		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	70504-2451	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Space Life Sciences-Flight NNH09ZTT002N
Start Date:	08/03/2010	**End Date:**	10/31/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: New end date is 10/31/2015 per NSSC information (Ed., 3/3/15) NOTE: New end date is 3/31/2015 per NSSC information (Ed., 1/22/14) NOTE: New end date is 12/31/13 per NSSC information (Ed., 8/6/13)		
Key Personnel Changes/Previous PI:	n/a		
COI Name (Institution):			
Grant/Contract No.:	NNX10AP91G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This proposal combines morphological measurements on micro-gravity germinated and grown (Brassica) roots that are exposed to high-gradient magnetic fields (HGMF) with a high-resolution spatial analysis of genes involved in gravitropism. Magnetic gradients exert a repulsive force on diamagnetic substances such as the starch in amyloplasts, which results in gravitropic-like curvature. The proposal is based on hardware that was developed for a previous flight experiment on STS-107. As in the originally proposed experiment, microscopic analysis of amyloplast size and distribution will be studied in the context of achieved curvature. In addition, high-resolution genetic profiling will be conducted that focuses on the cytoskeleton and starch development. The result of the proposed work will enable directional growth of seedlings and eventually mature plants under microgravity conditions using HGMF. Three conditions will be used to differentiate microgravity from uniform magnetic fields and HGMF conditions. Only HGMF is expected to influence root curvature. A uniform magnetic field will serve as control for possible secondary magnetic effects. This condition will be compared to an in-flight control without magnetic field to determine growth rate, gene expression, growth, and directionality (autotropism). Root material from these experiments will be used for investigations on cytoskeletal organization and the expression of starch-related genes. A second aspect of the proposed work will investigate the process of gravisensing, whether dependent on amyloplast position, movement, or force exerted onto amyloplasts. Space experimentation is necessary to establish a threshold value for a directional magnetic force. In addition to providing insight in the fundamental mechanism of the gravisensing system of plants, the data from this experiment can be used to develop technology to provide a directional, gravity-like stimulus to plants during early stages of germination. The proposed experiment can be performed with existing hardware, be adapted easily to changing flight schedules, and requires minimal crew time. The anticipated benefits include a better understanding of root development, effects of amyloplast displacement on root development and curvature, and compensatory measures to alleviate microgravity conditions. A newly developed gene extraction technology is ideally suited for gene-profiling studies and has far-reaching implications for studying gene expression, especially for confined space flight experiments. Additional benefits of the project include training of students at the undergraduate and graduate levels.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In addition to an improved understanding of basic biological questions, such as identification of the site of graviperception, curvature-induced (but not gravity-dependent) gene expression, the experiment also tests the effect of magnetic fields on plant roots. Therefore, biological questions are investigated in conjunction with physical principles. We intend to analyze the effect of static magnetic fields on biological systems. This work will provide a starting point for the current discussion of the detrimental effects of (electro)magnetic fields. In addition, understanding basic biological questions will improve the prospect for enhanced plant productivity, resource optimization, and reduced maintenance.
Task Progress:	Report for activities since April 2015 Time line: Space-X3 launched April 18, 2014; the payload returned May 18 and arrived at KSC 5/21/14. Post flight processing was accomplished between 5/22 and 5/25/14. Ground control experiments (performed Aug. 14, 2014) ran into additional delays because of a shortage of hardware (MFCs were badly corroded during flight experiment so that flight hardware could not be used for the ground control). The first ground control experiment resulted in lack of germination (likely the result of RNALater related NH3 contamination), and fixative leaked. Difficulties to obtain approval for a second clinorotated ground control delayed the second ground control experiment until Jan. 7, 2015. A no-cost extension ensured that most (but not all) analyses were completed. Subsequent work was substantially affected by the departure of the post-doctoral research associate who obtained a job offer in Korea that could not be refused or postponed. The analyses of cytoskeletal arrangements was performed by confocal microcopy by June 2015; analyses are ongoing and almost complete. This effort is part of Andrea Edge's PhD research. Light microscopy started with sample embedding during de-integration; sectioning was performed in a six-month period (4/2015 – 9/2015, together with corresponding samples from ground controls). Analyses are partially complete and ongoing. Substantial results were obtained. The experiment provided exciting insights in the gravitropic mechanism of roots, their adaptive responses, and gene regulation during flight and ground controls. Results: Flight experiment: A) Imaging. The most important part of the flight experiment was the confirmation of the effect of high-gradient magnetic fields (HGMF) on root curvature. The data provide clear evidence that HGMF does induce root curvature as theory and ground experiments predicted. Because curvature occurred in weightlessness conditions, the gravisensing system of roots is not compromised by space flight. The enhancement of the amyloplast size in space-grown roots (see D, below) indicates metabolic adaptation to reduced effective gravity levels. The induction of curvature by HGMF in space-grown roots indicates that amyloplasts are indeed the organelles that elicit differential growth and therefore the originators of gravitropism. B) Cytoskeletal arrangements. The comparison between static, clinorotated, and space grown roots did not show any substantial differences between the cytoskeletal organization of actin or microtubules. A high resolution analysis showed a greater variation in orientation and coherency of both microtubules and F-actin after clinorotation. C) Gene expression. The examined genes were selected to metabolic relevance and connection to differential elongation. The chosen genes were (in order of average transcription levels with accession number in parentheses): Cytochrome c oxidase, amylase, Ubiquitin, Phosphofructokinase, Hexokinase, Alcohol dehydrogenase, Tubulin, Actin 7, P Pin-formed 1, 3, 7, Sucrose synthase, IAA transcription factor 5, Glucokinase, Glucose-6-phosphate dehydrogenase, and Triacylglycerol lipase. Comparisons of different conditions included flight conditions, clinorotation, and static ground conditions, with and without HGMF. Measurements were performed in the root tip, root proper, root shoot junction, and seed tissue. Gene transcription levels were assessed with validated primers using SYBR-Green based qPCR analysis. Comparisons were based on regressions of the transcription levels of all examined genes and served as basis for the identification of up-or downregulated genes. The methodology resulted in a publication (Houston et al., 2015) and provides reliable assessment of gene regulation. Genes that responded to space condition, clinorotation, and HGMF included auxin transporter PIN1-7, enzymes involved in glycolysis (hexokinase, glucokinase), and sucrose synthase.

	However, not all tissues responded equally. D) Light microscopy. The purpose of these studies was the identification of amyloplast positioning and size in the root cap. The difference between the size of amyloplasts from space-grown roots and those obtained from clinorotated or static roots was derived from systematic measurements of all detectable starch grains. The difference was highly significant and amounted to an approximate size increase of 30% in diameter. The amyloplast studies are important because they indicate that clinorotation (i.e., hyper-stimulation) reduces amyloplast size. Therefore clinorotation is likely to reduce gravisensitivity. It is anticipated that a publication reporting on the results of the flight experiment will be submitted by summer 2016.
Bibliography Type:	Description: (Last Updated: 01/20/2026)
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